Problem 1) One in which the motion comes back to each position it has visited after a specified time with the same velocity.

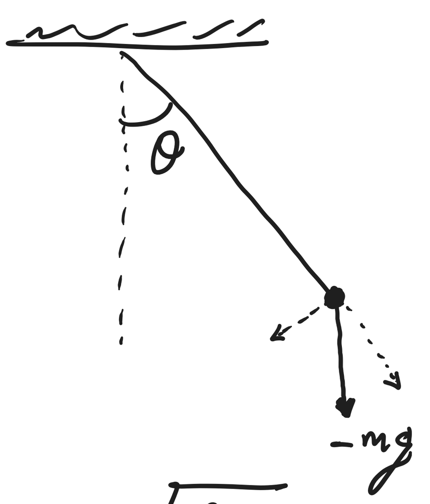
Problem 2) ① Simple harmonic, damped, driven-damped.

② $\ddot{x}$: acceleration ; $2\beta\dot{x}$: dissipative force $\omega_0^2 x$: restoring force

③ The factor 2 simplifies the final solution to the ODE and has no impact on the solution.

Problem 3) The tangential component of $-mg$ is,

$-mg\sin\theta = m(l\ddot{\theta})$



This eqn. is not analytically solvable.
We must make a small angle approximation.

$\lim_{\theta\to 0}\sin\theta = \theta$

$\Rightarrow \ddot{\theta} + \omega^2\theta \cong 0$; $\theta\to 0$ where $\omega = \sqrt{\frac{g}{l}}$

$\Rightarrow \theta(t) \cong A\cos(\omega t + \varphi)$ for small θ.

A and φ are unknown constants determined by the initial conditions of the problem.

Problem 4) ① A is the amplitude.

② $x(t) = A\sin(\omega t - \delta) \Rightarrow \dot{x}(t) = A\omega\cos(\omega t - \delta)$

$\Rightarrow \ddot{x}(t) = -A\omega^2\sin(\omega t - \delta)$

$\Rightarrow \boxed{\ddot{x} + \omega^2 x = 0}$ simple harmonic motion.

③ $\left.\begin{array}{l} x = A\sin(\omega t - \delta) \\ \dot{x} = A\omega\cos(\omega t - \delta) \end{array}\right\} \Rightarrow \left(\frac{x}{A}\right)^2 + \left(\frac{\dot{x}}{A\omega}\right)^2 = \sin^2(\cdot) + \cos^2(\cdot) = 1$

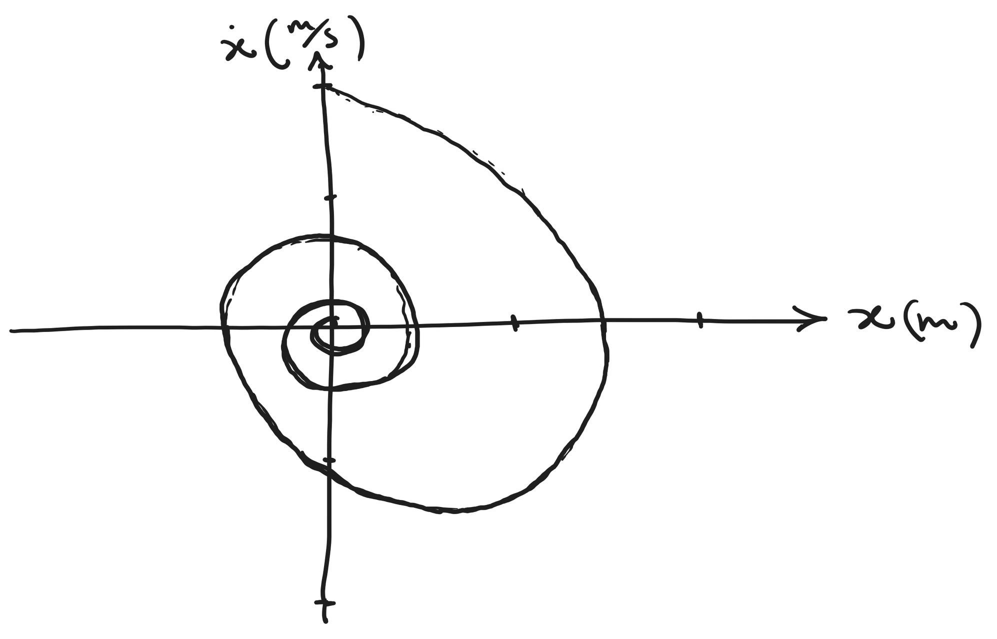
This is the eqn of an ellipsoid. ✓

Problem 5) ① Underdamped motion.

② $x(t) = Ae^{(-\frac{\beta}{2}t)}\sin(\omega t - \delta)$

$\Rightarrow \dot{x}(t) = -Ae^{-\beta t}\left[\beta\cos(\omega t - \delta) + \omega\sin(\omega t - \delta)\right]$; $T = \frac{1}{2}m\dot{x}^2$

③



Problem 6) $L = T - U$; The kinetic energy is composed of radial and tangential components,

$$T = \frac{1}{2}m\left(\dot{x}^2 + (\underbrace{(l+x)\dot{\theta}}_{\text{tangential speed}})^2\right) \quad \text{I}$$

The potential energy has gravity and spring components;

$$U(x,\theta) = -mg(l+x)\cos\theta + \frac{1}{2}kx^2 \quad \text{II}$$

I II $\Rightarrow L = \frac{1}{2}m\left(\dot{x}^2 + (l+x)^2\dot{\theta}^2\right) + mg(l+x)\cos\theta - \frac{1}{2}kx^2$

There are two variables here: (x, θ)

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow m\ddot{x} = m(l+x)\dot{\theta}^2 + mg\cos\theta - kx & \text{III} \\ \frac{\partial L}{\partial \theta} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow \frac{d}{dt}\left(m(l+x)^2\dot{\theta}\right) = -mg(l+x)\sin\theta & \text{IV} \end{cases}$$

$\Rightarrow m(l+x)^2\ddot{\theta} + 2m(l+x)\dot{x}\dot{\theta} = -mg(l+x)\sin\theta$

$\Rightarrow m(l+x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg\sin\theta$ V

III is simply the Newton's law $F = ma$ with the second term being related to the centripetal acceleration $-(l+x)\dot{\theta}^2$.

IV states that torque equals the rate of change of angular momentum.